Gayoung Kwak

AI & Data Science for Healthcare

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PROFILE

Healthcare AI and applied machine-learning researcher combining food and nutrition domain knowledge with software and data science. Experience with clinical records, public data, and medical images, with an emphasis on reproducible evaluation and practical limitations.

EXPERIENCE

Research Intern

Integrative Research Center for Cerebrovascular and Cardiovascular Diseases, Yonsei University College of Medicine | Jul 2024 - Jul 2025

- Led an end-to-end study of LLM-based clinical feature selection for ICU delirium prediction.
- Constructed a retrospective MIMIC-IV cohort of 32,574 first ICU stays, including 2,728 incident delirium cases, and mapped 107 features from 11 studies to 151 analysis variables.
- Compared four procedures: Scoring, Ranking, Sequencing, and log-prior across GPT-4.1, Claude Sonnet 4.5, and Gemini 2.5 Pro, with five independent runs per setting.
- The best 30-feature LLM subset retained 93.0% of the full-feature XGBoost AUROC, supporting the potential of LLMs as assistive tools for feature prioritisation in clinical prediction research.

RESEARCH

LLM-Based Feature Prioritisation for ICU Delirium Prediction: Reproducibility and Predictive Utility

First author | Under Review at Smart Health (Journal Impact Factor 3.4)

Evaluated LLMs as literature-informed priors for clinical feature selection using MIMIC-IV and downstream XGBoost benchmarking.

EDUCATION

B.S. Food Science and Nutrition

Kyung Hee University, Seoul Campus | 2022 - Present

B.S. Software Convergence (Data Science Track)

Kyung Hee University, Global Campus | 2024 - Present

Exchange - Data Science and Engineering

Instituto Superior Tecnico, University of Lisbon | Sep 2025 - Jan 2026

Coursework: Deep Learning, Natural Language, Information Systems and Databases, Parallel and Heterogeneous Computing Systems

HONORS

Grand Prize and Mayor of Seoul Award

2026 Seoul Metropolitan Government Big Data Competition | Startup Division, Product & Service Development

Awarded for Seoul Foodservice Site Intelligence, a service predicting three-year survival for prospective foodservice business locations in Seoul. Role: Data & ML Lead.

SKILLS

Languages Python, SQL, R, C++, JavaScript

ML / DL PyTorch, scikit-learn

LLM Tools Hugging Face, LangChain

Data Analysis NumPy, pandas

Web React

Cloud AWS, GCP

Version Control Git

LANGUAGES

Korean Native

English IELTS Academic 7.0 (C1) | Oct 31, 2026

SELECTED WORK

DentiScan

Data Analysis Capstone Design | 2026

Built an RF-DETR + Swin Transformer pipeline to filter false positives. Overall precision increased from 67.41% to 80.63%, while recall decreased from 73.25% to 62.74%. Deployed on AWS with lesion-level and mean confidence scores.

SS-TI

Food and Nutrition Capstone Design | 2024

Built a 40-item sugar-and-sodium web assessment, evaluated with KMO and Cronbach's alpha. Personalized feedback helps users understand their eating habits and identify practical dietary improvements.

Book-Delivery Robot Hubs

Seoul Gwanak-gu Youth Data Science Program | 2023

Combined nine public datasets and OLS regression to prioritize library robot hubs and routes.

Food Poisoning Battle

Kyung Hee University KHUthon | 2023

Designed the concept, rules, progression, and UI for a Unity strategy game based on pathogen growth conditions.